



Heat Transfer

Heat can be transferred from one body to another by means of conduction, convection or radiation. *Excel Review Center*

Thermal Conduction (or heat conduction) is the transfer of thermal energy between neighboring molecules in a substance due to a temperature gradient.

Fourier's Law (Law of Thermal Conduction)

"The time rate of heat transfer through a material is proportional to the gradient in the temperature and to the area, at right angle to the gradient, through which the heat is flowing."

The rate of heat transfer through conduction is:

$$\frac{Q}{t} = \frac{kA\Delta T}{L} = \frac{A\Delta T}{R}$$

where:

Excel Review Center

$\frac{Q}{t}$ = rate of heat conduction (power of conduction)

k = thermal conductivity

ΔT = change in temperature

L = thickness of slab or material

A = area

$\frac{\Delta T}{L}$ = temperature gradient

R = thermal resistance

Thermal conductivity, k, of the material is a measure of its ability to conduct heat. The larger the k, the faster the heat transfer.

Thermal resistance, R, is the insulating ability of a layer of material.

$$R = \frac{\text{thickness}}{\text{thermal conductivity}} = \frac{L}{k}$$

Convection is the transfer of heat by the motion of a volume of hot fluid from one place to another.

The rate of heat transfer through convection is:

$$\frac{Q}{t} = hA\Delta T$$

where:

Excel Review Center

$\frac{Q}{t}$ = rate of heat convection (power of convection)

h = convection coefficient

ΔT = change in temperature

A = area

Pass Math in board
exam easily. Subscribe
to MEME Review thru
Excel Online



Radiation This takes place by means of electromagnetic waves which require no material medium for their passage.

The rate of heat transfer through radiation at which an object of surface area A and absolute temperature T emits radiation is given by the Stefan-Boltzmann law as:

$$\frac{Q}{t} = \sigma eAT^4$$

$$P = \sigma eAT^4$$

$$P_{\text{net}} = \sigma eA(T_2^4 - T_1^4)$$

where:

$\sigma = 5.67 \times 10^{-8} \text{ W/m}^2$ (Stefan-Boltzmann constant)

e = emissivity (value from 0 to 1)

e = 0 for perfect reflector (body that does not radiate at all)

e = 1 for blackbody *Excel Review Center*

Emissivity is the ability of an object to emit radiation. A perfect radiator has emissivity of 1. Dark objects generally have high values of emissivity while light or shiny objects have relatively low values of emissivity. A **black body** is an idealized object that absorbs all electromagnetic radiation falling on it.

DID YOU KNOW?

The term "Black Body" was introduced in 1860 by Gustav Kirchhoff!

Internal Energy

The term "energy" is coined by Thomas Young in 1807. All potential and kinetic energies of the atoms and molecules in a substance are called internal energy.

Units of internal energy (U) are BTU/lb-mol and kJ/kmol. *Excel Review Center*

Units of specific internal energy (u) are BTU/lbm and kJ/kg.

All forms of thermodynamic energy are measured by BTU.

$$U = (MW)(u)$$

$$U = \pm Q \pm W$$

where:

+ Q = heat enters the system / heat applied to the system

- Q = heat leaves the system / heat removed from the system

+ W = work done on the system by the surroundings

- W = work done by the system to the surroundings

Ideal Gases

Excel Review Center

Ideal gas is one when kept at constant temperature, obeys the gas laws exactly.

Avogadro's law:

"Equal volumes of different gases at the same temperature and pressure contain equal number of molecules."

$$\frac{PV}{T} = R$$

$$PV = nRT$$

$$n = \frac{m}{MW}$$

where:

V = volume of gas

P = pressure of gas

T = absolute temperature

R = universal gas constant

n = number of moles

m = mass of substance

MW = molecular weight of substance

Density of Ideal Gases

$$PV = nRT$$

$$PV = \frac{m}{MW} RT$$

$$\rho = \frac{m}{V} = \frac{P(MW)}{RT}$$

Values of Universal Gas Constant, R:

R = 8.3143 kJ/kmol-K

R = 0.0821 atm-L/mol-K

R = 1.96 cal/mol-K

R = 8.314 J/mol-K

R = 82.1 atm-cm³/kmol-K

R = 0.0821 atm-m³/kmol-K

R = 8314.3 kg-m²/s²kmol-K

R = 8314.3 m³ Pa/kmol-K

R = 8.314 x 10⁷ erg/mol-K

R = 1545.33 ft-lbf/lbmol-°R

R = 1.986 Btu/lbmol-°R

R = 0.7302 atm-ft³/lbmol-°R

R = 10.73 ft³-lbf/in²-lbmol-°R

Excel Review Center

Kinetic Gas Theory

States that gases are made up of a large number of small particles (atoms or molecules), all of which are in constant, random motion.

The rapidly moving particles constantly collide with each other and with the walls of the container. This theory explains macroscopic properties of gases, such as pressure, temperature or volume by considering their molecular composition and motion. This also predicts the velocity distribution of gas molecules as a function of temperature.

Excel Review Center